Sea-Bird Scientific
13431 NE 20th Street
Bellevue, WA 98005
USA

+1 425-643-9866
seabird@seabird.com
www.seabird.com

SENSOR SERIAL NUMBER: 1640
CALIBRATION DATE: 12-Aug-26

SBE 39 PRESSURE CALIBRATION DATA
508 psia S/N 5733

COEFFICIENTS:

PA0 =	9.086472e-02	PTCA0 =	7.380118e+01
PA1 =	2.386052e-02	PTCA1 =	4.231137e-01
PA2 =	1.837972e-09	PTCA2 =	-1.115846e-02
PTHA0 =	-8.843057e+01	PTCB0 =	2.526988e+01
PTHA1 =	4.665743e-02	PTCB1 =	-2.025000e-03
PTHA2 =	-4.811046e-07	PTCB2 =	0.000000e+00

PRESSURE SPAN CALIBRATION / THERMAL CORRECTION

PRESSURE (PSIA)	INSTRUMENT OUTPUT (counts)	THERMISTOR OUTPUT (counts)	COMPUTED PRESSURE (PSIA)	RESIDUAL (%FSR)	TEMP (°C)	THERMISTOR OUTPUT (counts)	INSTRUMENT OUTPUT (counts)
14.51	680.8	2457.0	14.51	-0.00	-1.50	1900.40	684.40
104.77	4454.3	2461.0	104.75	-0.00	1.00	1956.30	685.43
204.79	8632.8	2462.0	204.74	-0.01	4.50	2034.40	686.62
304.79	12809.5	2463.0	304.76	-0.01	8.00	2112.80	687.68
404.81	16983.6	2463.0	404.77	-0.01	11.50	2191.10	688.43
504.80	21155.7	2463.0	504.80	-0.00	15.00	2270.00	688.98
404.75	16985.3	2462.0	404.81	0.01	18.50	2348.80	689.06
304.76	12811.0	2462.0	304.79	0.01	22.00	2427.70	688.92
204.75	8634.1	2462.0	204.77	0.01	25.50	2506.60	688.56
104.74	4455.1	2462.0	104.77	0.01	29.00	2585.80	687.69
14.49	679.8	2461.0	14.49	-0.00	32.50	2665.10	687.14

TEMPERATURE (°C)	SPAN
-5.00	25.28
35.00	25.20

y = thermistor output (counts)

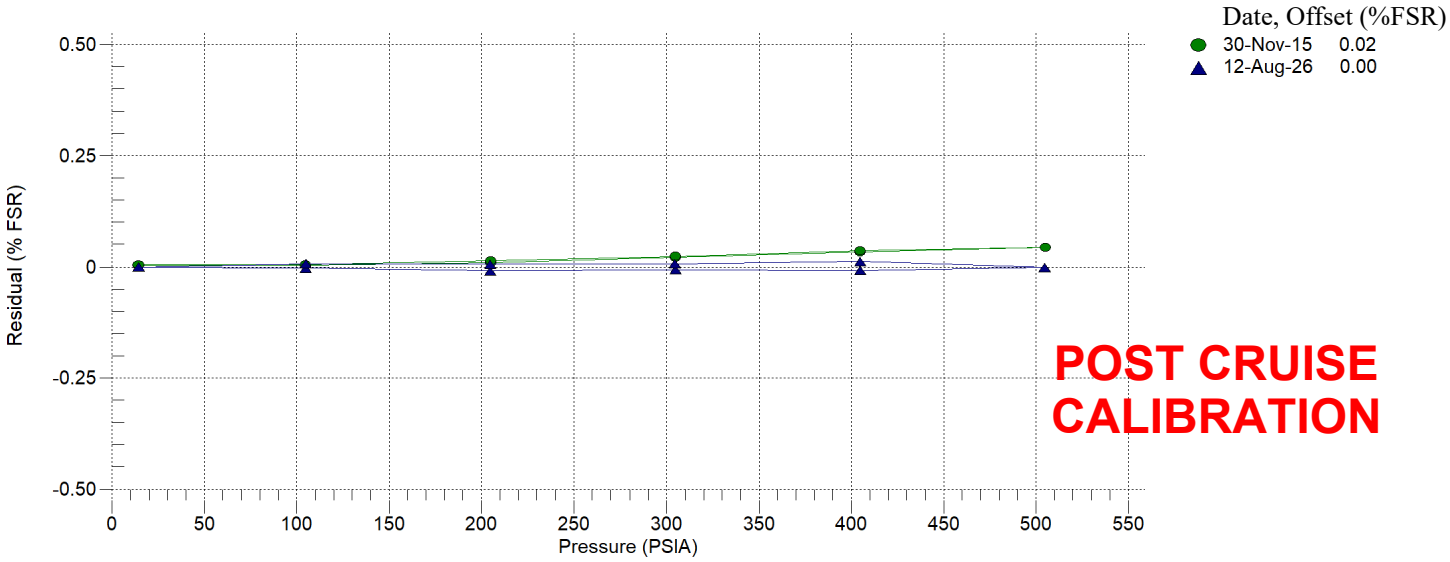
$t = PTHA0 + PTHA1 * y + PTHA2 * y^2$

$x = \text{instrument output} - PTCA0 - PTCA1 * t - PTCA2 * t^2$

$n = x * PTCB0 / (PTCB0 + PTCB1 * t + PTCB2 * t^2)$

$\text{pressure (PSIA)} = PA0 + PA1 * n + PA2 * n^2$

Residual (%FSR) = (computed pressure - true pressure) * 100 / Full Scale Range



POST CRUISE CALIBRATION